

College of Computing and Information Technology

Department of Information Technology

Database Systems - Midterm Exam

Course Code	CPIT320	Term	Spring 2026
Course	Database Systems	Duration	60 minutes
Exam Type	Midterm	Total Marks	20

Student Name: Student ID:

Section: Signature:

Instructions: Answer all questions. For multiple-choice items, select one answer only. Show concise reasoning where requested. Write SQL statements clearly.

Question 1 - Multiple Choice (6 marks)

Q1.1 Which relational-model concept uniquely identifies a row in a table? **[2 marks]**

- A) Primary key
- B) View
- C) Trigger
- D) Cursor

Q1.2 Which normal form removes partial dependency on a composite key? **[2 marks]**

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

Q1.3 Which SQL clause filters rows before grouping? **[2 marks]**

- A) WHERE
- B) HAVING
- C) ORDER BY
- D) UNION

Question 2 - Short Answer (6 marks)

Q2(a) Explain the difference between a primary key and a foreign key. **[3 marks]**

Q2(b) Given a relation with a partial dependency, explain how you would decompose it to satisfy 2NF. **[3 marks]**

Question 3 - SQL (8 marks)

Q3(a) Write an SQL SELECT statement that lists student names from STUDENT where GPA is greater than 4.0. **[4 marks]**

Q3(b) Write an SQL query that counts students in each department using GROUP BY. **[4 marks]**
